

Manoj Arulmurugan

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EDUCATION

MS, Data Science, University of Wisconsin-Madison
Statistical Modeling, Optimization, NLP, Advanced Deep Learning (GPA: 3.82/4.0)

Sep 2024 – May 2026 | Madison, WI

Graduate Certificate, Data Science and Applications, IIT Madras ☑

Jun 2023 – Dec 2023

B.Tech, Computer Science, Vellore Institute of Technology GPA: 8.96/10 ☑

Jul 2019 – May 2023 | Chennai, India

SKILLS

Languages & Frameworks: Python, R, SQL, PyTorch, PySpark, Apache Spark, XGBoost, Hugging Face Transformers

Cloud & MLOps: GCP (BigQuery, Dataproc, Cloud Run, GCS), Snowflake, MLflow, Docker, FastAPI, dbt, Git

Data & Tools: LangChain, LangGraph, Darts, Pandas, NumPy, Polars, Feature Stores, ETL, Tableau, PowerBI

ML & Statistics: Forecasting, MLOps, GenAI, LLMs, RAG, Transformers, Multi-Agent Systems, A/B & Hypothesis Testing

EXPERIENCE

Machine Learning Intern, Calix, Inc.

Jun 2025 – Dec 2025 | San Jose, CA, USA

- Architected a centralized **Feature Store** on Snowflake integrating **10+** data sources at account-week granularity, engineering **100+** features to standardize model inputs and eliminate redundant query development across the team.
- Redesigned the production Account-Level Regression model end-to-end, building **13 week-specific models** with algorithm-specific feature selection (**VIF + Genetic Algorithm** for linear; **SHAP** for tree models), achieving **~50% MAPE reduction**.
- Productionized the full pipeline on **Snowflake-dbt**, reducing dimensionality by **~80%** through Genetic Algorithm-based feature selection and optimizing the training window to meet the 30-minute warehouse timeout and reduce compute costs.
- Contributed to the broader **forecasting ensemble** (SARIMA + LSTM + ALR) achieving **~60%** overall accuracy improvement.
- Collaborated with Business Analysts and Data Engineers on **cohort-based account analysis** tracking revenue expansion and churn trends across customer acquisition, growth, and churn stages.

Operations Data Analyst, Shell India Market Pvt. Ltd. ☑

Aug 2023 – Aug 2024 | Bangalore, India

- Owned analytics for B2C apps, building **PowerBI** dashboards querying a **10TB data lake** via **SQL** across ServiceNow, Jira, and SharePoint; automated ETL pipelines, accelerating processing 3x and driving **\$250k in cost savings**.
- Designed KPI tracking framework adopted as primary analytics tool across B2C and B2B teams at Shell India; recognized with the **Shell Mobility Wall of Fame Award (Q1 2024)** ☑ for delivering high-impact data solutions.

Data Science Intern, Sedin Technologies Pvt. Ltd. ☑

Sep 2022 – Feb 2023 | Chennai, India

- Built a two-stage ML forecasting pipeline for an Indian fintech NBFC using **MLForecast**, engineering features across **400+** customer/region/risk combinations and applying clustering to reduce dimensionality to 6 behavioral groups.
- Improved forecast accuracy from **76% to 95%** using a **RandomForest + XGBoost** ensemble (SMAPE 4.5%, R^2 of 0.99), supporting 3-month loan disbursement forecasts for resource allocation and budgeting decisions.

PROJECTS

E-Commerce Recommendation System: Production-Scale GRU4Rec ☑, Big Data · GCP · MLOps

- Processed **280M raw events** (445K users, 284K items) using **PySpark** on Dataproc; applied bot removal, k=3 core filtering, and 30-min session boundary detection to produce **1.45M** training sessions aligned to **T4Rec** (RecSys 2021) evaluation protocol
- Trained GRU4Rec achieving **NDCG@20=0.26**, exceeding the T4Rec XLNet benchmark (NDCG@20=0.2546); **scaled to 1M users on Vertex AI (A100)**, deployed FastAPI on Cloud Run (<500ms), and built an automated **concept-drift retraining pipeline** via Cloud Scheduler triggering Vertex AI when rolling NDCG@20 dropped >15%

Customer Churn Prediction, Non-Contractual Setting ☑, Time-Series · Statistical Modeling · ML

- Engineered dynamic churn labels on a real enterprise sales dataset using per customer-product **Empirical Cumulative Distribution Function** thresholds, capturing behavioral heterogeneity across **370k** records without fixed cutoffs.
- Benchmarked forecasting-based, statistical, and ML-based churn detection approaches; **LightGBM** and **RandomForest** achieved best performance at 87% accuracy, outperforming time-series and distribution-based baselines.

Time-Aware RAG: Temporal Retrieval + Fast MRAG Re-Ranking ☑, Retrieval-Augmented Generation (RAG)

- Built a time-aware RAG pipeline for "as-of/when" QA on year-anchored FineWeb-Edu Q-passage pairs; fine-tuned **Contriever** with **temporal hard negatives** via triplet margin loss to reduce time-mismatched retrieval on CAQA and in-domain test sets.
- Built **MRAG re-ranker** via Sliding-Window MaxSim + temporal-decay fusion; **Hit@1 59%** vs 40.4% baseline, in-domain **84.9%**

SquadPlanner: AI-powered Trip Planning Application ☑, LangGraph · Multi-Agent Systems · MongoDB

- Architected a stateful multi-agent LLM system in **LangGraph**; MongoDBCheckpointner **serializes TripState** across HTTP for **human-in-the-loop** approval; post-gen edits re-enter graph at the affected node via astream_events, skipping upstream.
- Parsed free-text into typed hard constraints gating tool selection; streamed **real-time agent progress** via SSE.